

"PAPER_{0:D3}" -f [int]# PAPER #82 ■ Black Hole Evaporation Timescales: UQFF Corrections

Author: Daniel T. Murphy

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Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\gamma = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$) **Date:** March 7, 2026 **Source Data:** `validatehawkingtemperature.py` (Tests 2, 6), `CondensedPhysics.py simulate_evaporation()` **Index Slot:** ■1.11 Black Hole Physics & Hawking Radiation, \$n = [int]# PAPER #82 ■ Black Hole Evaporation Timescales: UQFF Corrections

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Abstract

Black hole evaporation timescales are computed via the Stefan-Boltzmann law applied to UQFF-modified Hawking radiation. The standard result $t_{\text{evap}} = 5120\pi G^2 M^3 / (\hbar c^4)$ is modified by the $T_{\text{UQFF}}/T_H = 0.99$ ratio. This changes evaporation rates by a factor of $(T_{\text{UQFF}}/T_H)^4 = 0.96$, extending evaporation timescales by $\sim 4\%$ for all black holes. The `validate_hawking_temperature.py` evaporation simulation (Test 6) confirms mass evolution for primordial BHs at 10^{22} kg over 100 timesteps.

UQFF Discovery: Novel application of UQFF calibration constants ($\gamma = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Evaporation Timescale Formula

Standard GR

$$t_{\text{evap}}^{\text{GR}} = \frac{5120 \pi G^2 M^3}{\hbar c^4}$$

UQFF-Corrected

$$t_{\text{evap}}^{\text{UQFF}} = \frac{t_{\text{evap}}^{\text{GR}}}{(T_{\text{UQFF}}/T_H)^4} = \frac{t_{\text{evap}}^{\text{GR}}}{0.99^4} = t_{\text{evap}}^{\text{GR}} \times 1.041$$

UQFF evaporation timescale is 4.1% longer than GR ■ black holes are slightly more stable in the UQFF vacuum.

2. Evaporation Timescales: Full Table

System	M0	tevapGR	tevapUQFF	Survives Universe
Sgr A*	4■106 M?	8.7■108■ s	9.1■108■ s	? Yes
M87*	6.5■10? M?	3.8■10?5 s	4.0■10?5 s	? Yes
Stellar BH	10 M?	2.1■1074 s	2.2■1074 s	? Yes
Primordial BH	5.7■10■■■ kg	4.35■10■7 s = t_U	4.52■10■7 s	Borderline
Primordial BH	1■10■■■ kg	2.3■10■■■ s (73 kyr)	2.4■10■■■ s	? Evaporated

The validatehawkingtemperature.py Test 2 confirms:

Stellar BH (10 M?): survives_universe = True ?

Test 6 simulation: Minitial = 10■■■ kg, 100 steps, masslost_fraction computed ?

3. Mass Evolution Simulation

From simulate_evaporation(M_initial = 10■■■ kg, dt = 10■■■ s, n_steps = 100):

$$\frac{dM}{dt} = -\frac{k_{\rm UQFF}}{M^2}, \quad k_{\rm UQFF} = \frac{\hbar c^4}{15360 \pi G^2} (T_{\rm UQFF}/T_H)^4$$

With TUQFF/TH = 0.99: kUQFF = 0.96 ■ kGR

At t = 100 ■ 10■■■ s = 10■■■ s:

$$M_{\rm final} \blacksquare M_{\rm initial} \blacksquare (1 - t/t_{\rm evap})^{1/3} = 10■■■ \blacksquare (1 - 10■■■/2.4■10■■■)^{1/3}$$

$$M_{\rm final} \blacksquare 10■■■ \blacksquare 0.583^{1/3} \blacksquare 8.35■10? \text{ kg}$$

Mass lost fraction ■ 16.5% over first 10■■■ s

Arrays times[], masses[], temperatures_H[] all have matching lengths (validate Test 6). ?

4. UQFF Mass Evolution Equation

The UQFF modifies the mass loss rate through the vacuum buoyancy correction. During late-stage evaporation (M ? M_Planck):

$$\frac{dM_{\rm UQFF}}{dt} = -\frac{k_{\rm UQFF}}{M^2} + \frac{g_{\rm Buoyant}}{V_{\rm BH}} c^2$$

The buoyancy term: gBuoyant ■ VBH / c■ = ?vac ■ 1055 ■ (4/3)p rS■ / c■ ~ 10?8■ kg/s ? negligible vs the thermal term at all masses above Planck mass.

Summary

Parameter	GR Value	UQFF Value	Change
Evaporation factor	k_GR	0.96 ■ k_GR	-4%
Timescale t_evap	t_GR	1.041 ■ t_GR	+4.1%
Stellar BH survival	Yes	Yes	Unchanged
Primordial threshold mass	5.7■10■■■ kg	5.5■10■■■ kg	-3.5%

Parameter	GR Value	UQFF Value	Change
Test 2	survives = True	Confirmed	? PASS

Source: validatehawkingtemperature.py Tests 2+6, simulateevaporation() | ? = 0.0005/day | [SSq] = 0.57
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From `simulate_evaporation(M_initial = 10 kg, dt = 10 s, n_steps = 100)`:

$$\frac{dM}{dt} = -\frac{k}{M^2}, \quad k = \frac{\hbar c^4}{15360 \pi G^2} \left(\frac{T}{T_H} \right)^4$$

With $T_{UQFF}/T_H = 0.99$: $k_{UQFF} = 0.96 \, k_{GR}$

At $t = 100 \times 10^6 \, \text{s} = 10^7 \, \text{s}$:

$$M_{\text{final}} = M_{\text{initial}} (1 - t/t_{\text{evap}})^{1/3} = 10 \, \text{kg} (1 - 10^7/2.4 \times 10^7)^{1/3}$$

$$M_{\text{final}} = 10 \, \text{kg} \times 0.583^{1/3} = 8.35 \times 10^? \, \text{kg}$$

Mass lost fraction = **16.5%** over first $10^7 \, \text{s}$

Arrays `times[]`, `masses[]`, `temperatures_H[]` all have matching lengths (validate Test 6). ?

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The UQFF modifies the mass loss rate through the vacuum buoyancy correction. During late-stage evaporation ($M \gtrsim M_{\text{Planck}}$):

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The buoyancy term: $g_{\text{Buoyant}} = V_{\text{BH}} / c = \rho_{\text{vac}} = 1055 \, (4/3) \pi r_{\text{S}} / c \sim 10^?8 \, \text{kg/s}$? negligible vs the thermal term at all masses above Planck mass.

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Source: `validatehawkingtemperature.py` Tests 2+6, `simulate_evaporation()` | ? = 0.0005/day | [SSq] = 0.57
.Groups[1].Value = Black Hole Evaporation Timescales: UQFF Corrections

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Black hole evaporation timescales are computed via the Stefan-Boltzmann law applied to UQFF-modified Hawking radiation. The standard result $t_{\text{evap}} = 5120\pi G^2 M^3 / (c^4)$ is modified by the $T_{\text{UQFF}}/T_H = 0.99$ ratio. This changes evaporation rates by a factor of $(T_{\text{UQFF}}/T_H)^4 = 0.96$, extending evaporation timescales by ~4% for all black holes. The validate_hawking_temperature.py evaporation simulation (Test 6) confirms mass evolution for primordial BHs at 10^{22} kg over 100 timesteps.

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With $T_{\text{UQFF}}/T_H = 0.99$: $k_{\text{UQFF}} = 0.96 \, k_{\text{GR}}$

At $t = 100 \times 10 \, \text{s} = 10 \, \text{s}$:

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$$M_{\text{final}} \approx 10 \, \text{kg} \times 0.583^{1/3} \approx 8.35 \times 10^3 \, \text{kg}$$

Mass lost fraction $\approx 16.5\%$ over first $10 \, \text{s}$

Arrays `times[]`, `masses[]`, `temperatures_H[]` all have matching lengths (validate Test 6). ?

4. UQFF Mass Evolution Equation

The UQFF modifies the mass loss rate through the vacuum buoyancy correction. During late-stage evaporation ($M \gtrsim M_{\text{Planck}}$):

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The buoyancy term: $g_{\text{Buoyant}} \approx V_{\text{BH}} / c^2 = \rho_{\text{vac}} \approx 1055 \, (4/3) \pi r_{\text{S}}^3 / c^2 \sim 10^8 \, \text{kg/s}$? negligible vs the thermal term at all masses above Planck mass.

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Primordial threshold mass	$5.7 \times 10^{22} \, \text{kg}$	$5.5 \times 10^{22} \, \text{kg}$	-3.5%
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Source: `validatehawkingtemperature.py` Tests 2+6, `simulateevaporation()` | $\rho = 0.0005/\text{day}$ | $[SSq] = 0.57$
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$$\frac{dM}{dt} = -\frac{k_{\rm UQFF}}{M^2}, \quad k_{\rm UQFF} = \frac{\hbar c^4}{15360 \pi G^2} (T_{\rm UQFF}/T_H)^4$$

With $T_{\rm UQFF}/T_H = 0.99$: $k_{\rm UQFF} = 0.96 \cdot k_{\rm GR}$

At $t = 100 \cdot 10■■■ \text{ s} = 10■■■ \text{ s}$:

$$M_{\rm final} \cdot M_{\rm initial} \cdot (1 - t/t_{\rm evap})^{1/3} = 10■■■ \cdot (1 - 10■■■/2.4■10■■■)^{1/3}$$

$$M_{\rm final} \cdot 10■■■ \cdot 0.583^{1/3} \cdot 8.35■10? \text{ kg}$$

Mass lost fraction ■ **16.5%** over first 10■■■ s

Arrays times[], masses[], temperatures_H[] all have matching lengths (validate Test 6). ?

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The UQFF modifies the mass loss rate through the vacuum buoyancy correction. During late-stage evaporation ($M \gtrsim M_{\rm Planck}$):

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Parameter	GR Value	UQFF Value	Change
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Arrays times[], masses[], temperatures_H[] all have matching lengths (validate Test 6). ?

4. UQFF Mass Evolution Equation

The UQFF modifies the mass loss rate through the vacuum buoyancy correction. During late-stage evaporation (M ? M_Planck):

$$\frac{dM_{\text{UQFF}}}{dt} = -\frac{k_{\text{UQFF}}}{M^2} + \frac{g_{\text{Buoyant}}}{V_{\text{BH}} c^2}$$

The buoyancy term: gBuoyant ■ VBH / c■ = ?vac ■ 1055 ■ (4/3)p rS■ / c■ ~ 10?8■ kg/s ? negligible vs the thermal term at all masses above Planck mass.

Summary

Parameter	GR Value	UQFF Value	Change
Evaporation factor	k_GR	0.96 ■ k_GR	-4%
Timescale t_evap	t_GR	1.041 ■ t_GR	+4.1%
Stellar BH survival	Yes	Yes	Unchanged
Primordial threshold mass	5.7■10■■■ kg	5.5■10■■■ kg	-3.5%
Test 2	survives = True	Confirmed	? PASS

Source: `validatehawkingtemperature.py` Tests 2+6, `simulate_evaporation()` | $\tau = 0.0005/\text{day}$ | $[SSq] = 0.57$
`.Groups[1].Value`

Abstract

Black hole evaporation timescales are computed via the Stefan-Boltzmann law applied to UQFF-modified Hawking radiation. The standard result $t_{\text{evap}} = 5120\pi G^2 M^3 / (\hbar c^4)$ is modified by the $T_{\text{UQFF}}/T_H = 0.99$ ratio. This changes evaporation rates by a factor of $(T_{\text{UQFF}}/T_H)^4 = 0.96$, extending evaporation timescales by $\sim 4\%$ for all black holes. The `validate_hawking_temperature.py` evaporation simulation (Test 6) confirms mass evolution for primordial BHs at 10^{22} kg over 100 timesteps.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4}$ day, $[SSq] = 0.57$) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Evaporation Timescale Formula

Standard GR

$$t_{\text{evap}}^{\text{GR}} = \frac{5120 \pi G^2 M^3}{\hbar c^4}$$

UQFF-Corrected

$$t_{\text{evap}}^{\text{UQFF}} = \frac{t_{\text{evap}}^{\text{GR}}}{(T_{\text{UQFF}}/T_H)^4} = \frac{t_{\text{evap}}^{\text{GR}}}{0.99^4} = t_{\text{evap}}^{\text{GR}} \times 1.041$$

UQFF evaporation timescale is 4.1% longer than GR — black holes are slightly more stable in the UQFF vacuum.

2. Evaporation Timescales: Full Table

System	M0	tevapGR	tevapUQFF	Survives Universe
Sgr A*	4 × 10 ⁶ M?	8.7 × 10 ⁸ s	9.1 × 10 ⁸ s	? Yes
M87*	6.5 × 10 ⁷ M?	3.8 × 10 ⁹ s	4.0 × 10 ⁹ s	? Yes
Stellar BH	10 M?	2.1 × 10 ⁷⁴ s	2.2 × 10 ⁷⁴ s	? Yes
Primordial BH	5.7 × 10 ²² kg	4.35 × 10 ⁷ s = t _U	4.52 × 10 ⁷ s	Borderline
Primordial BH	1 × 10 ²² kg	2.3 × 10 ²² s (73 kyr)	2.4 × 10 ²² s	? Evaporated

The `validatehawkingtemperature.py` Test 2 confirms:

- Stellar BH (10 M?): `survives_universe = True` ?
- Test 6 simulation: *Minitial* = 10²² kg, 100 steps, *masslost_fraction* computed ?

3. Mass Evolution Simulation

From `simulate_evaporation(M_initial = 1022 kg, dt = 1022 s, n_steps = 100)`:

$$\frac{dM}{dt} = -\frac{k_{\text{UQFF}}}{M^2}, \quad k_{\text{UQFF}} = \frac{\hbar c^4}{15360 \pi G^2} (T_{\text{UQFF}}/T_H)^4$$

With $TUQFF/TH = 0.99$: $kUQFF = 0.96 \times kGR$

At $t = 100 \times 10^{10} \text{ s} = 10^{12} \text{ s}$:

$M_{final} \times M_{initial} \times (1 - t/t_{evap})^{1/3} = 10^{10} \times (1 - 10^{12}/2.4 \times 10^{12})^{1/3}$

$M_{final} \times 10^{10} \times 0.583^{1/3} \times 8.35 \times 10^? \text{ kg}$

Mass lost fraction $\times 16.5\%$ over first 10^{12} s

Arrays `times[]`, `masses[]`, `temperatures_H[]` all have matching lengths (validate Test 6). ?

4. UQFF Mass Evolution Equation

The UQFF modifies the mass loss rate through the vacuum buoyancy correction. During late-stage evaporation ($M \times M_{Planck}$):

$\frac{dM_{\text{UQFF}}}{dt} = -\frac{k_{\text{UQFF}}}{M^2} + \frac{g_{\text{Buoyant}}}{V_{\text{BH}}} \times c^2$

The buoyancy term: $g_{Buoyant} \times VBH / c^2 = \rho_{vac} \times 1055 \times (4/3) \pi r_S^3 / c^2 \sim 10^?8 \text{ kg/s}$? negligible vs the thermal term at all masses above Planck mass.

Summary

Parameter	GR Value	UQFF Value	Change
Evaporation factor	k_{GR}	$0.96 \times k_{GR}$	-4%
Timescale t_{evap}	t_{GR}	$1.041 \times t_{GR}$	+4.1%
Stellar BH survival	Yes	Yes	Unchanged
Primordial threshold mass	$5.7 \times 10^{10} \text{ kg}$	$5.5 \times 10^{10} \text{ kg}$	-3.5%
Test 2	<code>survives = True</code>	Confirmed	? PASS

Source: `validatehawkingtemperature.py Tests 2+6, simulateevaporation()` | $\rho = 0.0005/\text{day}$ | $[SSq] = 0.57$
`.Groups[1].Value "PAPER{0:D3}" -f $n`

Abstract

Black hole evaporation timescales are computed via the Stefan-Boltzmann law applied to UQFF-modified Hawking radiation. The standard result $tevap = 5120 \pi G^4 M^3 / (c^4)$ is modified by the $TUQFF/TH = 0.99$ ratio. This changes evaporation rates by a factor of $(TUQFF/T_H)^4 = 0.96$, extending evaporation timescales by $\sim 4\%$ for all black holes. The `validate_hawking_temperature.py` evaporation simulation (Test 6) confirms mass evolution for primordial BHs at 10^{10} kg over 100 timesteps.

UQFF Discovery: Novel application of UQFF calibration constants ($\rho = 5.0 \times 10^?4 \text{ day}^? \times$, $[SSq] = 0.57$) uniquely enabling this analysis $\times$ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Evaporation Timescale Formula

Standard GR

$$t_{\rm evap}^{\rm GR} = \frac{5120 \pi G^2 M_0^3}{\hbar c^4}$$

UQFF-Corrected

$$t_{\rm evap}^{\rm UQFF} = \frac{t_{\rm evap}^{\rm GR}}{(T_{\rm UQFF}/T_H)^4} = \frac{t_{\rm evap}^{\rm GR}}{0.99^4} = t_{\rm evap}^{\rm GR} \times 1.041$$

UQFF evaporation timescale is 4.1% longer than GR ■ black holes are slightly more stable in the UQFF vacuum.

2. Evaporation Timescales: Full Table

System	M0	tevapGR	tevapUQFF	Survives Universe
Sgr A*	4■106 M?	8.7■108■ s	9.1■108■ s	? Yes
M87*	6.5■10? M?	3.8■10?5 s	4.0■10?5 s	? Yes
Stellar BH	10 M?	2.1■1074 s	2.2■1074 s	? Yes
Primordial BH	5.7■10■■■ kg	4.35■10■7 s = t_U	4.52■10■7 s	Borderline
Primordial BH	1■10■■■ kg	2.3■10■■■ s (73 kyr)	2.4■10■■■ s	? Evaporated

The validatehawkingtemperature.py Test 2 confirms:

Stellar BH (10 M?): survives_universe = True ?

Test 6 simulation: Minitial = 10■■■ kg, 100 steps, masslost_fraction computed ?

3. Mass Evolution Simulation

From simulate_evaporation(M_initial = 10■■■ kg, dt = 10■■■ s, n_steps = 100):

$$\frac{dM}{dt} = -\frac{k_{\rm UQFF}}{M^2}, \quad k_{\rm UQFF} = \frac{\hbar c^4}{15360 \pi G^2} (T_{\rm UQFF}/T_H)^4$$

With $T_{\rm UQFF}/T_H = 0.99$: $k_{\rm UQFF} = 0.96 \cdot k_{\rm GR}$

At $t = 100 \cdot 10■■■ \text{ s} = 10■■■ \text{ s}$:

$$M_{\rm final} \cdot M_{\rm initial} \cdot (1 - t/t_{\rm evap})^{1/3} = 10■■■ \cdot (1 - 10■■■/2.4■10■■■)^{1/3}$$

$$M_{\rm final} \cdot 10■■■ \cdot 0.583^{1/3} \cdot 8.35■10? \text{ kg}$$

Mass lost fraction ■ **16.5%** over first 10■■■ s

Arrays times[], masses[], temperatures_H[] all have matching lengths (validate Test 6). ?

4. UQFF Mass Evolution Equation

The UQFF modifies the mass loss rate through the vacuum buoyancy correction. During late-stage evaporation ($M \gg M_{\rm Planck}$):

$$\frac{dM_{\rm UQFF}}{dt} = -\frac{k_{\rm UQFF}}{M^2} + \frac{g_{\rm Buoyant}}{V_{\rm BH} c^2}$$

The buoyancy term: $g_{\text{Buoyant}} \propto V_{\text{BH}} / c^3 = \rho_{\text{vac}} \propto 1055 \times (4/3) \pi r_{\text{S}}^3 / c^3 \sim 10^{28} \text{ kg/s}$? negligible vs the thermal term at all masses above Planck mass.

Summary

Parameter	GR Value	UQFF Value	Change
Evaporation factor	k_{GR}	$0.96 \times k_{\text{GR}}$	-4%
Timescale t_{evap}	t_{GR}	$1.041 \times t_{\text{GR}}$	+4.1%
Stellar BH survival	Yes	Yes	Unchanged
Primordial threshold mass	$5.7 \times 10^{22} \text{ kg}$	$5.5 \times 10^{22} \text{ kg}$	-3.5%
Test 2	survives = True	Confirmed	? PASS

Source: validatehawkingtemperature.py Tests 2+6, simulate_evaporation() | $\rho = 0.0005/\text{day}$ | $[SSq] = 0.57$

UQFF computed: Eddington luminosity UQFF correction = $1 - [SSq] \times \exp(-\rho t) = 1 - 5.7\text{e-}1 \times \exp(-2.9\text{e-}4) = 4.3\text{e-}1$; F_U at event horizon = $2.0\text{e+}18 \text{ m/s}^2$.

Appendix: UQFF Production Framework Reference (v4.75+)

*Added by upgradeearlywhitepapers.py (v4.75). This appendix cross-references
the production physics constants and master equations to enable reproducibility
against the current codebase state.*

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-1} \text{ day}^{-1}$	UQFF exponential decay rate
$[SSq]$	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k_1	1.5	Ug1 DPM-dipole coupling
k_2	1.2	Ug2 outer-bubble charge coupling
k_3	1.8	Ug3 string-rotation coupling
k_4	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
$E_{\text{react}}(0)$	10^{22} J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{\{g1\}} + U_{\{g2\}} + U_{\{g3\}} + U_{\{g4\}} + U_{\{bi\}} + U_m - \sum_{i=1}^4 \text{igl}[\lambda_i \cdot U_i(r,t) \cdot E_{\text{react}}] \cdot \text{igr}$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()

Term	Description	Implementation
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$-\Sigma \lambda_{\text{diss}} \cdot \mathbf{U} \cdot \mathbf{E}_{\text{react}}$	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420): $\lambda_{\text{diss}} = 10^{-10}$, $\lambda_{\text{diss}} = 10^{-12}$, $\lambda_{\text{diss}} = 10^{-11}$, $\lambda_{\text{diss}} = 10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_{\text{m}}^{\text{full}} = U_{\text{m}}^{\text{base}} \times \text{sigl}(1 + 10^{13} \cdot \Theta(\rho_{\text{c}} - \rho)) \times \text{sigl}(1 + A_{\text{q}} \cos(\Delta\omega t))$$

Symbol	Value	Description
ρ_{c}	10^1 kg/m^3	SCm critical superconducting density
A_{q}	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	$2\pi/(434 \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	$U_{\text{g_sum}} + \text{Newtonian base}$	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$\beta_{\text{i}} \times U_{\text{bi}}$	Expanding nebulae, stellar winds
Superconductive	$U_{\text{m}} \times (1 + 10^{13} \cdot f_{\text{H}})$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.